UNIT I : Introduction to Finite Element Methods

1.1 What is the Finite Element Method?

FEM is a **numerical technique** to find approximate solutions of differential / integral equations that describe physical **field problems** (displacement, temperature, pressure, potential ...) on complicated geometries, loads and boundary conditions where exact (analytical) solutions are not possible.

- The structure (continuum) is divided into many small pieces called **finite elements**, joined at points called **nodes**.
- Over each element the unknown field is approximated by a simple polynomial (interpolation / shape functions).
- Element equations are written, **assembled** into global equations and solved for nodal unknowns.

$$[K]\{Q\} = \{F\}$$

where $[K]$ = global stiffness matrix, $\{Q\}$ = nodal displacement vector, $\{F\}$ = global load vector.

Steps in FEM (general procedure)

1. **Discretisation** : divide the body into elements (type, size, number).
2. Select **displacement (interpolation) function** $u = [N]\{q\}$.
3. Derive **element stiffness matrix** $[k]$ and element load vector $\{f\}$.
4. **Assemble** element equations to get global equations $[K]\{Q\} = \{F\}$.
5. Apply **boundary conditions** (supports) and solve for $\{Q\}$.
6. Post-process : strains $\boldsymbol{\varepsilon} = [B]\{q\}$, stresses $\boldsymbol{\sigma} = [D]\{\boldsymbol{\varepsilon}\}$, reactions.

Applications of FEM

- **Structural** : stress analysis of beams, trusses, frames, plates, shells, machine parts, bridges, aircraft & automobile bodies.
- **Thermal** : conduction, convection in fins, walls, engine blocks, heat exchangers.
- **Fluid flow** : seepage, potential flow, lubrication, aerodynamics.
- **Dynamics** : vibration, modal analysis, crash and impact. **Others** : electromagnetics, biomechanics, acoustics.

Advantages: *Advantages : handles irregular geometry, any loading, mixed materials, any boundary condition, gives stress at every point. Disadvantage : results are approximate; needs a computer and engineering judgement (mesh quality).*

1.2 Stress and Equilibrium

Consider a small element of a body. The internal forces per unit area are ***stresses****. In 3-D there are 6 independent stress components (because of complementary shear,* $\tau_{xy}=\tau_{yx}$ *etc.).*

3-D: 6 components

$\{\sigma\}=[\sigma x\ \sigma y\ \sigma z\ \tau xy\ \tau yz\ \tau zx]^T$

2-D stress element

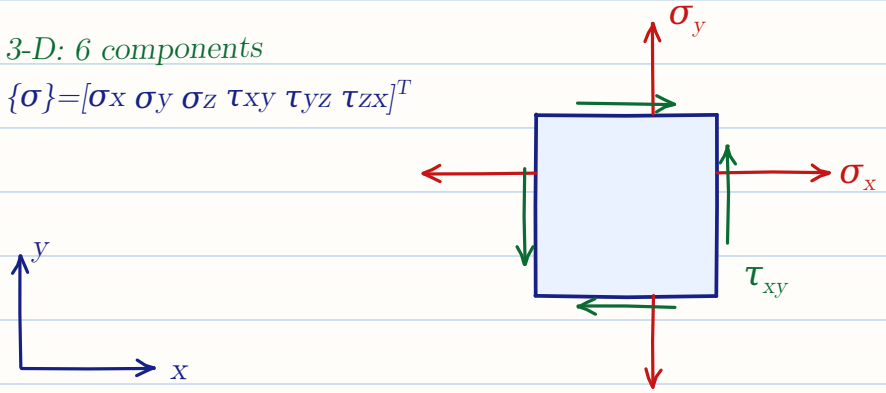


Fig 1.1 Stress components on an element

$$\{\sigma\} = [\sigma_x\ \sigma_y\ \sigma_z\ \tau_{yz}\ \tau_{xz}\ \tau_{xy}]^T$$

Equilibrium equations (3-D)

Take an elemental volume dx dy dz with body force per unit volume X, Y, Z (e.g. weight, centrifugal force). Sum of forces = 0 in each direction :

$$\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y + \partial\tau_{xz}/\partial z + X = 0$$

$$\partial\tau_{xy}/\partial x + \partial\sigma_y/\partial y + \partial\tau_{yz}/\partial z + Y = 0$$

$$\partial\tau_{xz}/\partial x + \partial\tau_{yz}/\partial y + \partial\sigma_z/\partial z + Z = 0$$

For ***2-D*** *problems :* $\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y + X = 0$ *and* $\partial\tau_{xy}/\partial x + \partial\sigma_y/\partial y + Y = 0$.

For ***1-D*** *:* $d\sigma/dx + f = 0$.

Boundary conditions

- ***Essential (geometric / Dirichlet)*** *BC : displacement is specified, e.g.* $u = 0$ *at a fixed support. Must be satisfied by the trial function.*
- ***Natural (force / Neumann)*** *BC : traction / force specified on the boundary, e.g.* $\sigma A = P$ *at a free end. Automatically satisfied in the weak (energy) form.*

Traction on an inclined boundary with direction cosines (n_x, n_y) *:* $T_x = \sigma_x n_x + \tau_{xy} n_y$, $T_y = \tau_{xy} n_x + \sigma_y n_y$.

1.3 Strain - Displacement Relations

Let u, v, w be the displacements in x, y, z. For small deformations (linear theory) :

$$\{\varepsilon\} = \begin{bmatrix} \partial u/\partial x \\ \partial v/\partial y \\ \partial w/\partial z \\ \partial v/\partial z + \partial w/\partial y \\ \partial u/\partial z + \partial w/\partial x \\ \partial u/\partial y + \partial v/\partial x \end{bmatrix}$$

i.e. $\varepsilon_x = \partial u/\partial x$, $\varepsilon_y = \partial v/\partial y$, $\varepsilon_z = \partial w/\partial z$ *(normal strains) and* γ_{yz}, γ_{xz}, γ_{xy} *(engineering shear strains).*

2-D *:* $\{\varepsilon\} = [\varepsilon_x \ \varepsilon_y \ \gamma_{xy}]^T = [\partial u/\partial x, \partial v/\partial y, \partial u/\partial y + \partial v/\partial x]^T$. ***1-D*** *:* $\varepsilon = du/dx$.

In FEM : $u = [N]\{q\} \Rightarrow \boldsymbol{\{\varepsilon\} = [B]\{q\}}$, *where [B] = strain-displacement matrix (derivatives of shape functions).*

1.4 Stress - Strain Relations (Hooke's law)

For a linear elastic isotropic material $\{\sigma\} = [D]\{\varepsilon\}$, *E = Young's modulus,* ν *= Poisson's ratio,* $G = E/2(1+\nu)$.

3-D elasticity matrix

$$[D] = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & (1-2\nu)/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & (1-2\nu)/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & (1-2\nu)/2 \end{bmatrix}$$

2-D elasticity matrices

$$\text{Plane STRESS } [D] = \frac{E}{(1-\nu^2)} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}$$

$$\text{Plane STRAIN } [D] = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & (1-2\nu)/2 \end{bmatrix}$$

- ***Plane stress*** *: thin plate loaded in its plane,* $\sigma_z = \tau_{xz} = \tau_{yz} = 0$ *(but* $\varepsilon_z \neq 0$*). e.g. thin plates, discs.*
- ***Plane strain*** *: long body with uniform section & load,* $\varepsilon_z = \gamma_{xz} = \gamma_{yz} = 0$ *(but* $\sigma_z \neq 0$*). e.g. dam, tunnel, long cylinder.*

1-D *:* $\sigma = E\,\varepsilon$*. Plane strain is obtained from plane stress by replacing* $E \rightarrow E/(1-\nu^2)$ *and* $\nu \rightarrow \nu/(1-\nu)$*.*

1.5 Potential Energy and Equilibrium

Total potential energy *Π = Strain energy (U) − Work done by external loads (W).*

$$\Pi = U - W = \tfrac{1}{2} \int \{\varepsilon\}^T\{\sigma\}\, dV - \int \{u\}^T\{X\}\, dV - \int \{u\}^T\{T\}\, dS - \Sigma\, \{u_i\}^T\{P_i\}$$

Principle: *Principle of Minimum Potential Energy : Among all admissible displacement fields (satisfying the essential BC) the one that satisfies equilibrium makes* Π *a minimum* $\Rightarrow \delta\Pi = 0$*, i.e.* $\partial\Pi/\partial q_i = 0$*.*

For a discretised system $\Pi = \tfrac{1}{2}\{Q\}^T[K]\{Q\} - \{Q\}^T\{F\}$*. Setting* $\partial\Pi/\partial\{Q\} = 0$ *gives* $\boldsymbol{[K]\{Q\} = \{F\}}$*, the equilibrium equation of FEM.*

Rayleigh - Ritz method

Assume the displacement as a finite series with unknown constants, $u = a_1\varphi_1(x) + a_2\varphi_2(x) + \ldots + a_n\varphi_n(x)$, each φ_i satisfying the essential BC. Substitute in Π and apply $\partial\Pi/\partial a_i = 0$ to get n algebraic equations.

Example : Bar of length L, constant AE, fixed at $x = 0$, axial load P at $x = L$. Take $u = a\,x$:

$$\Pi = \tfrac{1}{2}\int_0^L AE\,(du/dx)^2\,dx - P\,u(L) = \tfrac{1}{2}\,AE\,a^2\,L - P\,a\,L$$

$$\partial\Pi/\partial a = AE\,a\,L - P\,L = 0 \Rightarrow a = P/AE,\ u = P\,x/AE \text{ (exact)}$$

- Limitation : trial function must be written for the **whole** domain, difficult for complex shapes. ***FEM = Rayleigh-Ritz applied piece-wise*** on every element with local shape functions.
- Galerkin method (weighted residual) : weight functions = shape functions; gives the same equations for structural problems.

1.6 Formulation of Finite Element Equations

Within an element $\{u\} = [N]\{q\}$, $\{\varepsilon\} = [B]\{q\}$, $\{\sigma\} = [D][B]\{q\}$. Substituting in Π (element level) :

$$\Pi_e = \tfrac{1}{2}\{q\}^T\left[\int [B]^T[D][B]\,dV\right]\{q\} - \{q\}^T\left(\int [N]^T\{X\}\,dV + \int [N]^T\{T\}\,dS + \{P\}\right)$$

$$[k] = \int_V [B]^T\,[D]\,[B]\,dV \quad \{f\} = \int_V [N]^T\{X\}dV + \int_S [N]^T\{T\}dS + \{P\}$$

Minimising Π_e gives the element equation $\mathbf{[k]\{q\} = \{f\}}$. Properties of $[k]$: (i) symmetric, (ii) singular (rigid-body motion possible before supports are applied), (iii) diagonal terms positive, (iv) banded.

1.7 One-Dimensional Problems : the Bar Element

A 1-D bar carries only axial load. Each node has **one** degree of freedom (axial displacement). A 2-noded linear element has 2 dof : $\{q\} = [q_1\ q_2]^T$.

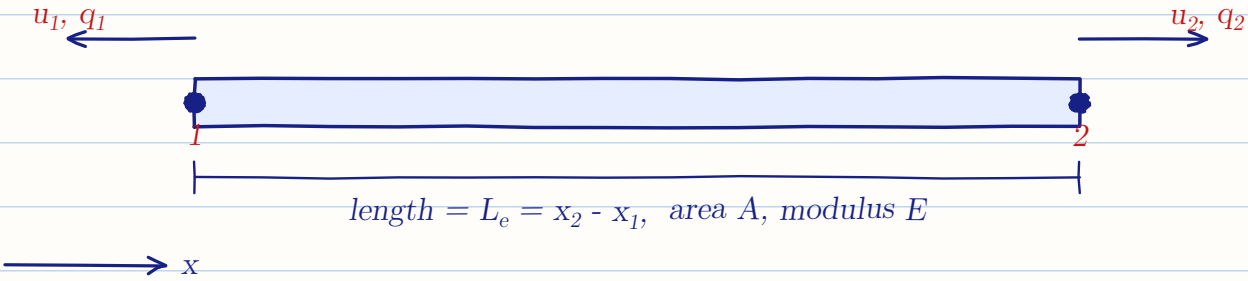


Fig 1.2 Two-noded 1-D bar element

Coordinates and shape functions

Use the ***natural (intrinsic) coordinate*** ξ *that goes from* -1 *at node 1 to* $+1$ *at node 2 :*

$$\xi = 2(x - x_1)/(x_2 - x_1) - 1,\ x = \tfrac{1}{2}(1-\xi)x_1 + \tfrac{1}{2}(1+\xi)x_2$$

$$N_1 = (1 - \xi)/2\ N_2 = (1 + \xi)/2\ u = N_1 q_1 + N_2 q_2$$

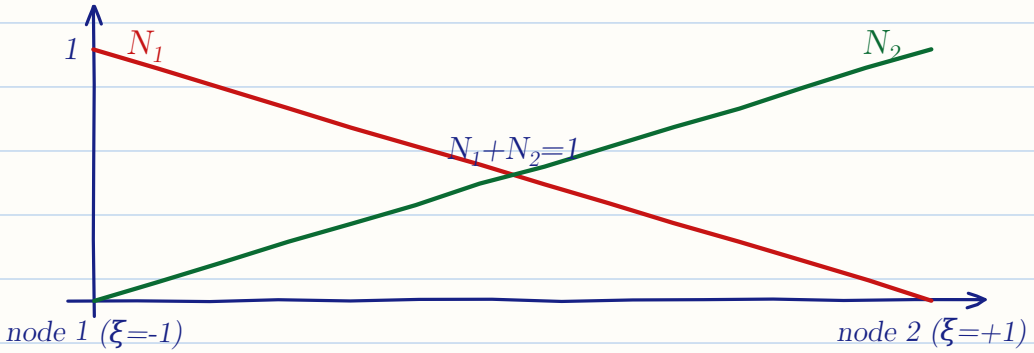


Fig 1.3 Linear shape functions

Properties : $N_i = 1$ *at node* i *and* 0 *at the other node;* $\Sigma N_i = 1$ *at every point (rigid-body motion is represented);* $dx = (L_e/2)\, d\xi$ *(Jacobian* $J = L_e/2$*).*

Strain-displacement matrix and stiffness matrix

$$\varepsilon = du/dx = (du/d\xi)(d\xi/dx) = (2/L_e)\,(du/d\xi) \Rightarrow [B] = (1/L_e)\,[-1\ \ 1]$$

$$[k] = \int [B]^T E\,[B]\,A\,dx = A_e E_e\, L_e/L_e^{\,2}\,[\ \ldots\ ]$$

$$[k] = A_e E_e / L_e \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Meaning: *Physical meaning : the bar behaves as a spring of stiffness* $k = AE/L$*. Column 1 = forces needed for unit displacement of node 1 with node 2 fixed.*

Load vectors

$$\text{Body force } f \text{ per unit volume}: \quad \{f_b\} = A_e L_e f/2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{Traction } T \text{ per unit area (on surface)}: \quad \{f_T\} = T\, L_e/2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{Thermal load (rise } \Delta T): \quad \{f_{th}\} = E_e A_e \alpha\, \Delta T \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

1.8 Requirements for Convergence; Interpolation Functions

For the FE solution to converge to the exact solution as the mesh is refined, the displacement (interpolation) function must satisfy :

1. ***Completeness*** : must include ***rigid-body modes*** and ***constant-strain states*** (so that when element size $\rightarrow 0$ the strain becomes uniform).
2. ***Compatibility (conformity)*** : displacement must be continuous across element boundaries (no gaps or overlaps). Elements satisfying it are called conforming.
3. ***Geometric isotropy (invariance)*** : the polynomial should not prefer a direction; the result must not depend on orientation of the axes (use all terms of Pascal triangle symmetrically).
4. Number of terms in polynomial = number of element dof.

Interpolation function (polynomial) is preferred because it is easy to differentiate and integrate and the order can be increased to improve accuracy. A 1-D polynomial : $u = \alpha_1 + \alpha_2 x + \alpha_3 x^2 + \ldots$

Pascal's triangle

Used to pick polynomial terms for 2-D elements so that the polynomial stays complete and symmetric in x and y :

constant			1			CST uses 1, x, y
linear			x		y	
quadratic		x^2		xy		y^2 Q4 adds xy
cubic		x^3		x^2y	xy^2	y^3
quartic	x^4		x^3y		x^2y^2	xy^3 y^4

Fig 1.4 Pascal's triangle

- *3 nodes (CST) : $u = \alpha_1 + \alpha_2 x + \alpha_3 y$ (complete linear, 3 terms). 4 nodes (Q4) : add the xy term (bilinear). 6 nodes (LST) : complete quadratic (6 terms).*

1.9 Assembly of Global Stiffness Matrix and Load Vector

*Consider a bar with n elements and n+1 nodes, global dof Q_1 ... Q_{n+1}. Element 1 connects nodes (1,2), element 2 connects (2,3), and so on. The element matrix is added to the global matrix at the rows & columns of its connected nodes (**direct stiffness method**). For 3 elements (4 nodes) :*

$$[K]\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1+k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2+k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix}\begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \end{Bmatrix}\begin{Bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{Bmatrix}$$

where $k_i = A_i E_i / L_i$. The global load vector is assembled in the same way : $F_2 = f_2^{(1)} + f_1^{(2)} +$ (point load at node 2), etc.

Treatment of boundary conditions

Before solving, [K] is singular. Supports must be applied. Two methods :

***(a) Elimination method** : for the dof $Q_r = 0$, strike out row r and column r (reduced system). Solve for the remaining Q. Then support reaction $R_r = [K]_{r,*}\{Q\} - F_r$.*

For a non-zero support displacement $Q_r = a$, move the column terms to the right side : $F_i \leftarrow F_i - K_{ir}\, a$ for all remaining rows.

***(b) Penalty approach** : add a very large number C ($\approx 10^4 \times \max K_{ii}$) to the diagonal K_{rr} and add $C \cdot a$ to F_r. Then solve the full system. Reaction $R_r = -C$*

$(Q_r - a)$.

Solved Example 1 : stepped bar

Bar fixed at node 1, $P = 50$ kN at node 3. $E = 200$ GPa $= 2\times10^5$ N/mm^2. Element 1: A_1=400 mm^2, L_1=300 mm; element 2: A_2=250 mm^2, L_2=300 mm. Find displacements, stresses, reaction.

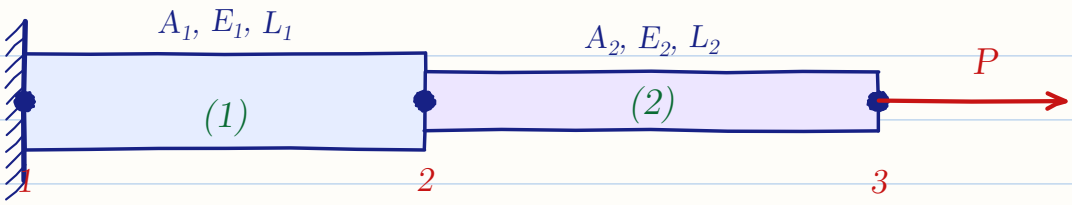


Fig 1.5

$k_1 = A_1E/L_1 = 400\times2\times10^5/300 = 2.667\times10^5$ N/mm ; $k_2 = 250\times2\times10^5/300 = 1.667\times10^5$ N/mm

$$[K]\begin{bmatrix} k_1 & -k_1 & 0 \\ -k_1 & k_1+k_2 & -k_2 \\ 0 & -k_2 & k_2 \end{bmatrix}\begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{Bmatrix} = [R_1,\ 0,\ 50000]^T$$

Apply $Q_1 = 0$ (strike row 1 & column 1). Reduced equations :

$$\begin{bmatrix} 4.333\times10^5 & -1.667\times10^5 \\ -1.667\times10^5 & 1.667\times10^5 \end{bmatrix}\begin{Bmatrix} Q_2 \\ Q_3 \end{Bmatrix} = [0,\ 50000]^T$$

Solving : $Q_2 = 0.1875$ mm, $Q_3 = 0.4875$ mm. Stresses : $\sigma_1 = E(Q_2-Q_1)/L_1 = 2\times10^5\times0.1875/300 = 125$ N/mm^2 ; $\sigma_2 = E(Q_3-Q_2)/L_2 = 2\times10^5\times0.3/300 = 200$ N/mm^2.

Reaction $R_1 = -k_1Q_2 = -2.667\times10^5\times0.1875 = -50\,000$ N (balances P). Check : $\sigma = P/A$ gives 125 and 200 OK.

1.10 Quadratic Shape Functions (3-noded bar)

A 3-noded element (nodes 1, 2 at the ends, node 3 at mid-length) gives a **parabolic** displacement and linearly varying strain : $u = \alpha_1 + \alpha_2\xi + \alpha_3\xi^2$. Using the condition $N_i(\text{node } j) = \delta_{ij}$:

$$N_1 = \xi(\xi - 1)/2 \quad N_2 = \xi(\xi + 1)/2 \quad N_3 = 1 - \xi^2$$

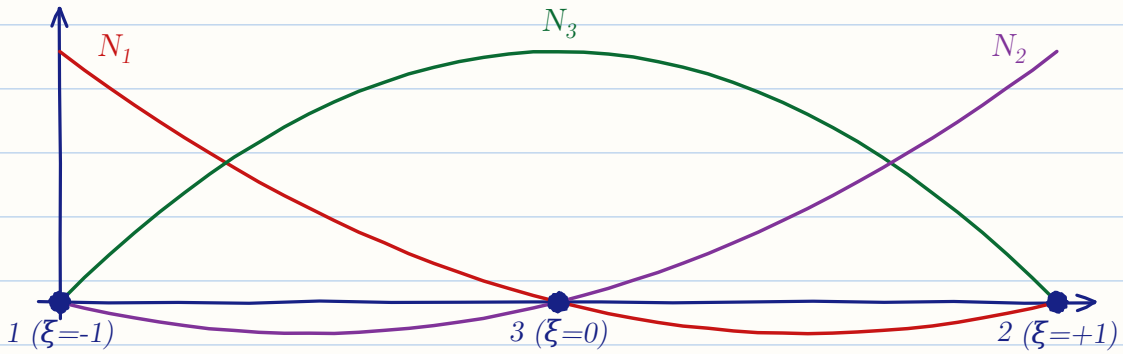


Fig 1.6 Quadratic shape functions (ξ = -1, +1, 0)

$[B] = (2/L_e)\ [\ dN_1/d\xi\ dN_2/d\xi\ dN_3/d\xi\ ] = (2/L_e)\ [\ (2\xi-1)/2\ ,\ (2\xi+1)/2\ ,\ -2\xi\ ]$.

Using $[k] = \int_{-1}^{1} [B]^T E[B]\ A\ (L_e/2)\ d\xi$:

$$[k] = A_e E_e/(3L_e) \begin{bmatrix} 7 & 1 & -8 \\ 1 & 7 & -8 \\ -8 & -8 & 16 \end{bmatrix}$$

$$\text{Body-force vector } \{f\} = f\ A_e\ L_e/6 \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}$$

Check : the sum of each row of $[k]$ is zero (rigid-body motion), and body-force weights 1/6, 1/6, 4/6 add to 1 (Simpson's rule).

Compare: Linear element : constant strain, needs many elements. Quadratic element : linear strain, better accuracy with fewer elements (same dof count gives higher accuracy).

1.11 More Solved Problems

Example 2 : Penalty approach (spring-like support)

Same stepped bar as Example 1. Use penalty method for $Q_1 = 0$. Choose $C = 10^4 \times \max K_{ii} \approx 4.3\times10^9$ N/mm.

$$\begin{bmatrix} 4.333\times10^5+C & -2.667\times10^5 & 0 \\ -2.667\times10^5 & 4.333\times10^5 & -1.667\times10^5 \\ 0 & -1.667\times10^5 & 1.667\times10^5 \end{bmatrix} \begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{Bmatrix} = [0,\ 0,\ 50000]^T$$

(Here $K_{11} = k_1 + C$.) Because C is huge, $Q_1 \approx 0$ (about 10^{-5} mm) and the remaining equations reduce to those of the elimination method, giving $Q_2 = 0.1875$ mm, $Q_3 = 0.4875$ mm. Reaction $R_1 = -C\,Q_1 \approx -50\,000$ N.

Remark: Penalty method keeps the matrix size unchanged and is easy to program; but too large C causes round-off errors, too small C gives an inexact constraint.

Example 3 : Bar under self weight (body force)

A uniform bar of length $L = 2$ m, $A = 100$ mm^2, $E = 2\times10^5$ N/mm^2, weight density $\rho g = 7.85\times10^{-5}$ N/mm^3, hangs from a fixed support at the top. Model with 2 elements of $L_e = 1000$ mm.

Body-force vector of each element : $f\,A\,L_e/2 = 7.85\times10^{-5}\times100\times1000/2 = 3.925$ N at each end node. Global load : $F_1 = R_1 + 3.925$, $F_2 = 7.85$, $F_3 = 3.925$.

$k = AE/L_e = 100\times2\times10^5/1000 = 2\times10^4$ N/mm. Apply $Q_1 = 0$:

$$\begin{bmatrix} 2k & -k \\ -k & k \end{bmatrix} \begin{Bmatrix} Q_2 \\ Q_3 \end{Bmatrix} = [7.85,\ 3.925]^T$$

Adding the equations: $k\,Q_2 = 11.775 \Rightarrow Q_2 = 5.89\times10^{-4}$ mm; then $k(Q_3-Q_2) = 3.925 \Rightarrow Q_3 = 7.85\times10^{-4}$ mm. Exact value at the tip $\rho g L^2/2E = 7.85\times10^{-4}$ mm, hence **exact at the nodes** (for linear elements and uniform loading).

Example 4 : Thermal stress in a bar

Steel bar ($E = 2\times10^5$ N/mm^2, $\alpha = 12\times10^{-6}$/°C, $A = 500$ mm^2, $L = 400$ mm) is rigidly fixed at both ends and heated by $\Delta T = 50$ °C. Find stress.

One element. Thermal load $\{f_{th}\} = EA\alpha\Delta T\,[-1,\ 1]^T = 2\times10^5\times500\times12\times10^{-6}\times50 = 60\,000$ N. Both ends fixed ($Q_1 = Q_2 = 0$), so all displacements are zero and the strain due to loading is zero. Mechanical strain $= -\alpha\Delta T$:

$\sigma = E(\varepsilon - \alpha\Delta T) = 2\times10^5(0 - 12\times10^{-6}\times50) = -120\ N/mm^2$ (compressive)

Example 5 : Quadratic element

A 3-noded bar ($L_e = 100$ mm, $A = 200\ mm^2$, $E = 2\times10^5\ N/mm^2$) is fixed at node 1; a load of 10 kN acts at node 2 (end). Find the displacement at the mid-node. $AE/(3L_e) = 200\times2\times10^5/300 = 1.333\times10^5$. With $Q_1 = 0$ the remaining equations (nodes 2, 3) are :

$$1.333\times10^5 \begin{bmatrix} 7 & -8 \\ -8 & 16 \end{bmatrix} \begin{Bmatrix} Q_2 \\ Q_3 \end{Bmatrix} = [10000,\ 0]^T$$

From row 2 : $Q_3 = Q_2/2$; row 1 : $7Q_2 - 4Q_2 = 3Q_2 = 10000/1.333\times10^5 = 0.075$ $\Rightarrow Q_2 = 0.025$ mm, $Q_3 = 0.0125$ mm. This equals the exact $PL/AE = 10^4\times100/(200\times2\times10^5) = 0.025$ mm.

Galerkin (weighted residual) form for the 1-D bar

Governing equation : $d/dx(AE\ du/dx) + f = 0$. Multiply by a weight function w ($= \delta u$, same as shape function) and integrate over the length; integrate by parts :

$$\int (dw/dx)\ AE\ (du/dx)\ dx = \int w\ f\ dx + [w\ AE\ du/dx]_{boundary}$$

Substituting $u = [N]\{q\}$, $w = [N]\{\delta q\}$ gives $\int [B]^T AE[B]\ dx\ \{q\} = \int [N]^T f\ dx +$ boundary forces, i.e. exactly $[k]\{q\} = \{f\}$. The natural BC appears automatically in the boundary term, hence it is called the **weak form**.

Shape function derivation (step by step, 2-noded)

1. Assume $u = \alpha_1 + \alpha_2 x$. 2. Apply $u(x_1) = q_1$, $u(x_2) = q_2$ and solve for α_1, α_2.
3. $\alpha_2 = (q_2 - q_1)/(x_2 - x_1)$; $\alpha_1 = (q_1x_2 - q_2x_1)/(x_2 - x_1)$.
4. Substitute back : $u = [(x_2-x)/L_e]\ q_1 + [(x-x_1)/L_e]\ q_2 \Rightarrow N_1 = (x_2-x)/L_e$, $N_2 = (x-x_1)/L_e$ (same as $(1\mp\xi)/2$).

Quick revision / expected questions

- Define FEM; list steps and applications. Write equilibrium equations in 2-D / 3-D.
- Derive strain-displacement and [D] for plane stress and plane strain. Differences?
- State principle of minimum potential energy; explain Rayleigh-Ritz with example.
- Derive shape functions, [B] and [k] of 1-D bar; assemble for a stepped bar;

elimination vs penalty.

- Convergence criteria; Pascal's triangle; derive quadratic shape functions.